

Got it — let's clear this up carefully. The distinction between **cointegration** and **correlation** comes down to:

1. What variables you apply them to (levels vs. returns).
 2. What kind of relationship they capture (short-run vs. long-run).
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1. Correlation

- **Definition:** Correlation measures the *short-run linear co-movement* between two series.
 - **Application:**
 - In finance, we almost always compute **correlation on returns**, not on price levels.
 - Why?
 - Price levels are typically **non-stationary** (they wander over time). If you compute correlation on two trending series, you can get misleadingly high values even if the series are unrelated ("spurious correlation").
 - Returns, on the other hand, are typically **stationary** (mean-reverting around zero), so correlation is meaningful.
 - **Interpretation:** Correlation tells us whether two assets tend to move together day-to-day (or week-to-week) in relative percentage changes.
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2. Cointegration

- **Definition:** Cointegration is a **long-run equilibrium relationship** between two (or more) *non-stationary* series.
 - **Application:**
 - Here we look at **price levels** (not returns).
 - If two assets are cointegrated, it means that although each one may wander (like a random walk), a particular linear combination of their price levels is **stationary**.
 - Example: Oil company stocks (XOM, CVX) both follow random walks individually, but their spread ($XOM - \beta \cdot CVX$) may revert around zero.
 - **Interpretation:** Cointegration tells us there is a *stable long-run relationship* — prices don't drift arbitrarily far apart, even if their short-term returns are uncorrelated.
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3. Why this distinction matters

- **Correlation:** Used with returns → measures short-term co-movement.
- **Cointegration:** Used with levels → measures long-term linkage.
- If we incorrectly applied correlation to price levels, we'd likely conclude "everything is highly correlated," which is misleading.
- If we incorrectly tested cointegration on returns, we'd almost always find no cointegration, since returns are already stationary (cointegration only makes sense for non-stationary series).